

# Muhammed Rayan

Computer Science Engineer

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## PROFESSIONAL SUMMARY

Solutions-driven Computer Science student focused on developing robust systems using clean engineering practices. Experienced in full-stack development, machine learning, and agile methodologies, with a proven ability to solve complex problems and deliver reliable software.

## EDUCATION

**College of Engineering Kidangoor** | 2022 - 2026  
*B.Tech in Computer Science and Engineering* | CGPA: 8.42  
**Razi International School** | 2019 - 2021  
*Higher Secondary (Science)* | 93.2%

## TECHNICAL SKILLS

**Languages:** JavaScript, TypeScript, Python, C/C++, SQL, HTML/CSS  
**Web Development:** React.js, Next.js, Node.js, REST APIs, HTML/CSS, MongoDB  
**ML, Deep Learning and Image Processing:** Machine Learning, Deep Learning, PyTorch, TensorFlow, Scikit-learn, OpenCV, SimpleITK, pydicom, DICOM  
**DevOps and Tools:** Git, CI/CD, Testing/QA, Syncfusion, Documentation  
**Engineering Practices:** SDLC, System Design, Agile, Scrum, Code Reviews

## EXPERIENCE

**Software Engineer Intern** Oct 2025 - Present  
*Auctapace Technologies* Remote  

- Building and enhancing Contract Lifecycle Management (CLM) features using React, Node.js, and reusable front-end components.
- Integrated REST API responses into front-end screens to populate contract data, workflow status, user actions, and dashboard views.
- Implemented feature updates, defect fixes, form validations, and UI improvements across agile sprint cycles.
- Tested features through manual QA, debugging, and review feedback to improve reliability before delivery.
- Prepared technical documentation, API notes, and implementation references for smoother handoff and team collaboration.

## PROJECTS

**InterLink - Event Platform** Jan 2025 - Mar 2025  
*React, TypeScript, MongoDB, JWT*  

- Designed a premium UI for event discovery and booking with React, TypeScript, and MongoDB.
- Built an Organizer Dashboard for event publishing, participant tracking, and ticket verification.
- Secured the platform with JWT authentication and optimized schemas for real-time analytics.

**Intracranial Aneurysm Detection** Aug 2025 - Ongoing  
*Python, PyTorch, nnU-Net, SimpleITK, pydicom, DICOM, MedGemma*  

- Developed a 3D full-resolution nnU-Net (PyTorch) for spatial localization and multi-label binary segmentation of intracranial aneurysms across 13 distinct anatomical zones.
- Engineered robust 3D medical image preprocessing pipelines using SimpleITK and pydicom to handle dynamic Hounsfield Unit (HU) normalization, orientation correction, and voxel-spacing alignment.
- Architected a memory-efficient inference and evaluation pipeline capable of processing 4,300+ DICOM series directly from compressed archives, computing real-world metrics without disk-write overhead.
- Designed a comparative model analysis framework to benchmark the primary CNN model against MedGemma, analyzing bounding box accuracy and classification performance.

**CLM User Dashboard** Dec 2025  
*React, Node.js, Syncfusion*  

- Built a role-based contract tracking dashboard with real-time status updates and workflow management.
- Implemented bulk assignment and approval workflows, advanced filtering, and infinite scroll for contract navigation.
- Developed a version history panel with document comparison using Syncfusion's Track Changes integration.

## CERTIFICATIONS

HackerRank SQL (Advanced) | Kaggle Machine Learning | Deloitte Data Analytics Simulation

## LANGUAGES

English (Conversational) | Malayalam (Native) | Arabic (Literate)